

```
class IShape
{
    virtual double area() const = 0;
    virtual double perimeter() const = 0;
    virtual void draw()
    {
        // WIN GDI - API
        // Disappearing rectangular outline
    }
};
```

```
class IList
{
    // user defined function which is not a special function
    // which is justifiably virtual here
    virtual ~IList()
    {
    }
};
```

```
class C
{
    public:
        const char* own_type()
        {
            return typeid(*this).name();
        }
};
```

```
#-----
Collectable.hpp
class Collectable
{
    public:
        virtual const char* typename() = 0;
        virtual size_t size() = 0;
};
```

IList.hpp

class IList

{

public:

virtual void insert_start(Collectable* pData) = 0;

virtual void insert_end(Collectable* pData) = 0;

virtual void show(Collectable* pData) = 0;

};

class Node

{

public:

Collectable* pData;

Node* prev;

Node* next;

Node(Collectable* _pData = 0);

};

CList.hpp

class CList : public IList

{

private:

Node* head_node;

const char* containee_type;

public:

CList(const char* containee_type);

void insert_start(Collectable* pData)

{

if(typeid(*pData).name() != containee_type)

throw type_error("bad type for data");

generic_insert(head_node, new Node(pData), head_node->next);

}

};

class Employee : public Collectable

{

public:

const char* type_name() const

{

return typeid(Employee).name();

}

```

size_t size() const
{
    return sizeof(Employee);
}
};

```

Client:

```

CList L(typeid(Employee).name());

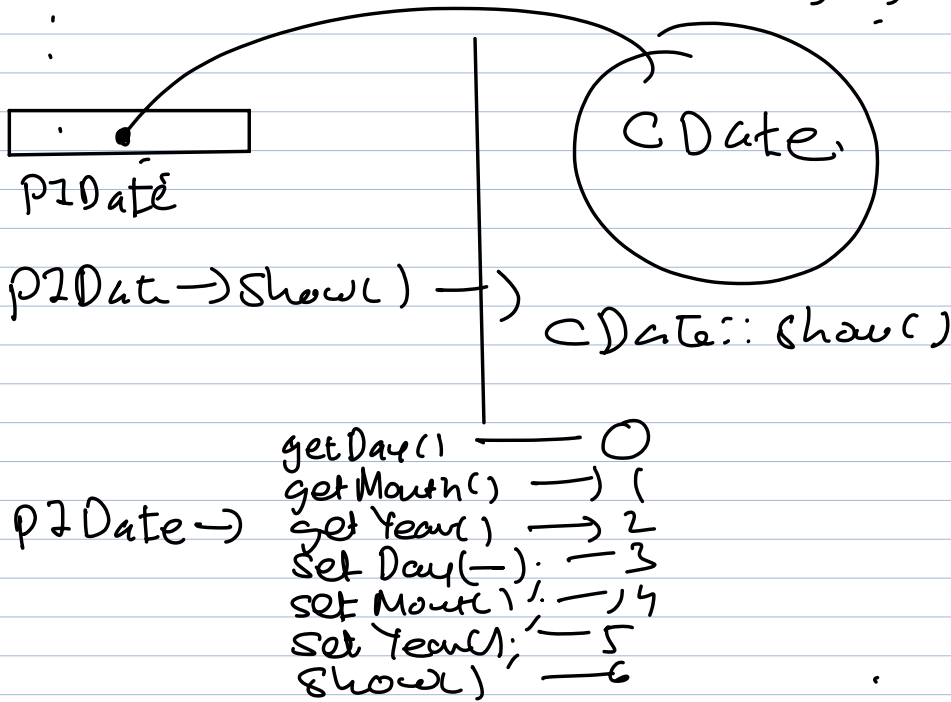
```

```

L.insert_start(new Employee());

```

$\text{IDate}^* \text{pIDate} = \text{IDate}::\text{getInstance}(\text{1, 2, 2028});$
 C++



6	show
5	setY
4	setM
3	setD
2	setY
1	setM.
0	setD
	Vtable.



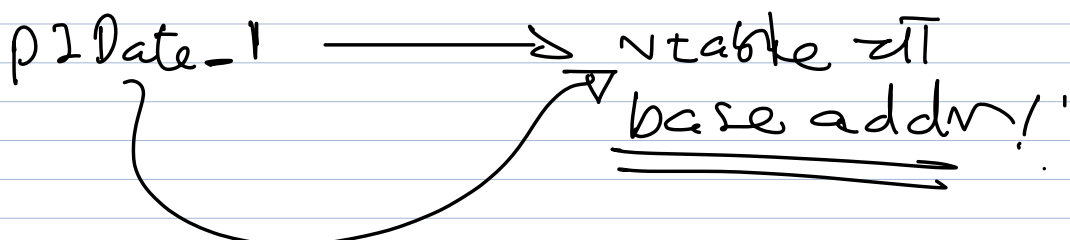
$p1Date \rightarrow \text{setMonth}(4);$

- ① push 4 on stack
- ② push Addr. in $p1Date$ on stack
- ③ Transfer control flow to $vtable[4];$

$IDate^* p1Date_1 = IDate::getInstance(-1, -1);$

$IDate^* p1Date_2 = IDate::getInstance(-1, -1);$

- $p1Date_1 \rightarrow \text{setMonth}(9);$
 $p1Date_2 \rightarrow \text{setMonth}(12);$
- | | |
|--------------------|--------------------|
| ① push 9 | ① push 12 |
| ② push $p1Date_1$ | ② push $p1Date_2$ |
| ③ call $vtable[4]$ | ③ call $vtable[4]$ |

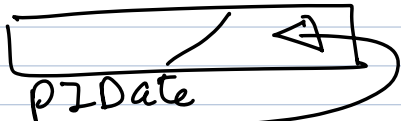


CDate → class → compile.

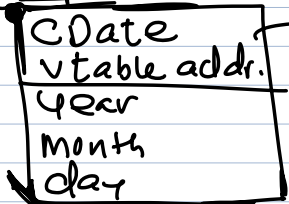
CDate



IDate | 7 virtual function.



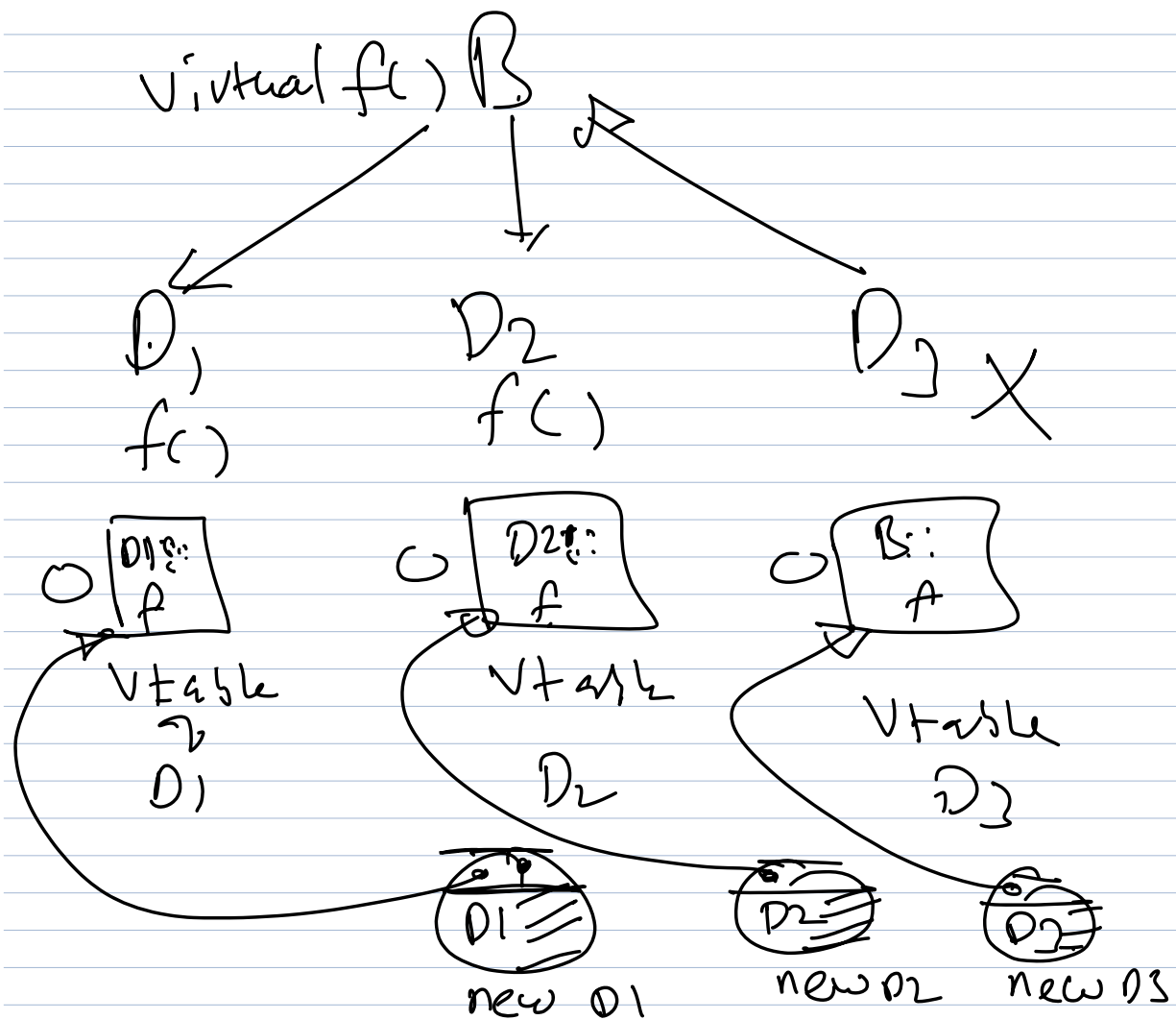
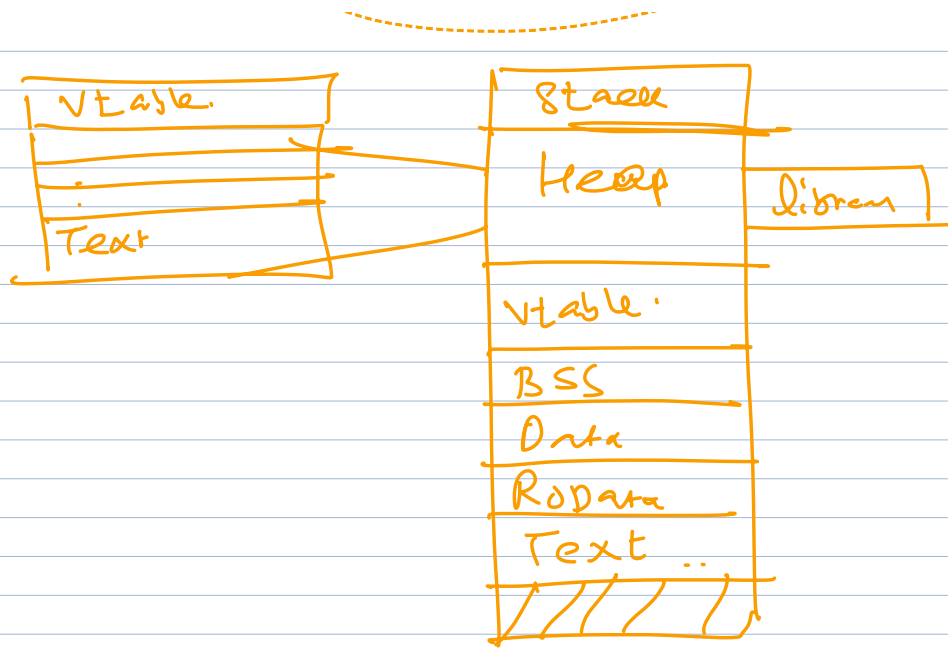
vptr.



& CDate::Show()
& CDate::SetYear()
& CDate::SetMonth()
& CDate::SetDay()
& CDate::getYear()
& CDate::getMonth()
& CDate::getDay()

Text section

get Year.	
Set Day	Set Month
Show()	Set Year
get Day	get Month



pBase